1. Definitions

① $a \equiv b \pmod{n}$ means $n \mid a-b$

i.e a is congruent to b modulo n

② if $0 \le b < n$ then b is residue of a

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congruence relation is equivalence equation

$\Rightarrow$ residue class

$\Rightarrow$ complete set of residue e.g $\{1, 2, \dots, n\} \pmod{n}$

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if $a \equiv a'$ and $b \equiv b' \pmod{n}$

then ① $a+b \equiv a'+b'$

② $a-b \equiv a'-b'$

③ $ab \equiv a'b'$ $\quad \because (a-a')b + a'(b-b') \equiv 0$

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④ polynomial f

with integer coefficients $\Rightarrow f(a) \equiv f(a')$

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① $d \mid n$ then $\cancel{d}a \equiv \cancel{d}a' \pmod{n} \iff a \equiv a' \pmod{n/d}$

② $(k, n) = 1$ then $\cancel{k}a \equiv \cancel{k}a' \pmod{n} \iff a \equiv a' \pmod{n}$

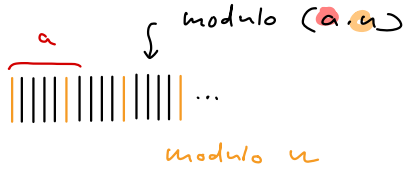
$$ka \equiv ka' \pmod{n} \iff \frac{k}{(k,n)} a \equiv \frac{k'}{(k,n)} a' \pmod{n/(k,n)}$$

$$\iff a \equiv a' \pmod{n/(k,n)}$$

2. Chinese Remainder Thm.

$ax \equiv b \pmod{n}$ is solvable iff $(a,n) \mid b$

$\Rightarrow$ $\cancel{a}^{\,0}x \equiv b \pmod{(a,n)}$



$\Leftarrow$ $a\underbrace{x}_{d \text{ solutions}} \equiv b \pmod{n}$ iff $\underbrace{a'}_{\text{reduced set of residue}}\underbrace{x}_{\text{unique solution}} \equiv b' \pmod{n'}$ i.e divided by $d = (a,n)$

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if ① p is prime then $ax \equiv b \pmod{p}$ i.e $aa^{-1} \equiv 1 \pmod{p}$

② $p \nmid a$ is always solvable.

Note. residues $0, 1, \ldots, p-1$

form field under add & mul $\pmod{p}$

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① $n = \overbrace{n_1 \cdots n_k}^{\text{coprime in pairs}}$

② $x \equiv c_j \pmod{n_j}$

CRT says we have unique solution

$$x \equiv \underbrace{m_1}_{n/n_1} m_1^{-1} c_1 + \cdots + \underbrace{m_k}_{n/n_k} m_k^{-1} c_k \pmod{n}$$

proof of uniqueness : $n_j \mid (x-y)$ therefore $n \mid (x-y)$

$x \equiv 2 \pmod{5}$ then

$x \equiv 3 \pmod{7}$

$x \equiv 4 \pmod{11}$

① CRT $x = 77x_1 + 55x_2 + 35x_3 = -18$

② check $2x_1 \equiv 2 \pmod{5} \rightarrow x_1 = 1$

$-x_2 \equiv 3 \pmod{7} \rightarrow x_2 = -3$

$2x_3 \equiv 4 \pmod{11} \rightarrow x_3 = 2$

3. Fermat and Euler's Thm.

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reduced set of residues

$\Rightarrow$ $\phi(n)$ elements with ① $0 \le a < n$

② $(a, n) = 1$

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multiplicative : $(n, n') = 1$ then $\phi(n)\phi(n') = \phi(nn')$

Claim bijection between $\{(a, a')\}$ and $\{b\}$

$(a, n) = 1$ $\quad$ $(b, nn') = 1$

$(a', n') = 1$

$\Rightarrow$ $an' + a'n$ is relatively prime with n, n', nn'

$\Leftarrow$ ① $mn' + m'n = 1$

therefore $(m, n) = 1$ and $(m', n') = 1$

② $bmn' + bm'n = b$

therefore $(bm, n) = 1$ and $(bm', n') = 1$

Fermat : prime p then $a^p \equiv a$

Euler : $(a, n) = 1$ then $a^{\phi(n)} \equiv 1 \pmod{n}$

$\because$ $\prod(ax) \equiv \prod(x)$ with (ax) and (x) runs through reduced set of residues

4. Willson's Thm.

$\underbrace{(p-1)!}_{\text{reduced set of residues}} \equiv -1 \pmod{p}$ prime

- $a\,a^{-1} \equiv 1 \pmod{p}$

  with $a \equiv a^{-1} \iff a \equiv \pm 1 \pmod{p}$

- $1 \cdot 2 \cdots (p-2) \cdot (p-1) \equiv p-1 \pmod{p}$

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if $(n-1)! \equiv -1 \pmod{n}$ then integer $n > 1$ is prime

because otherwise $n \mid (n-1)!$

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equation $x^2 \equiv -1 \pmod{p}$ with $r = (p-1)/2$

① if prime $p \equiv 1 \pmod{4}$ then $x = \pm(r!)$

because $\underbrace{(1) \cdots (r)}_{\equiv r!}\,\underbrace{(-r) \cdots (-1)}_{\equiv r!} \equiv -1 \pmod{p}$

② if prime $p \equiv 3 \pmod{4}$ then no solution

otherwise $x^{p-1} \equiv x^{2r} \equiv (-1)^r \equiv (-1) \pmod{p}$

$\rightarrow\leftarrow$ (Fermat)

5. Lagrange Thm.

$f(x)$ polynomial with $\Rightarrow$ $f(x) \equiv 0 \pmod{p}$ has at most n solutions

prime

(i) degree n

(ii) integer coefficients

(iii) coefficient of x^n not divisible by p

$\because$ $f(x) - \cancel{f(a)} \equiv (x-a)\underbrace{g(x)}_{\text{at most } n-1 \text{ zeros}} \pmod{p}$

$f(x) \equiv g(x) \pmod{p}$ means coefficients of like powers are congruent.

$f(x) \equiv 0$ has full complement $a_1, \ldots, a_n$ of solutions $\pmod{p}$

$\Rightarrow$ $f(x) \equiv c(x-a_1)\cdots(x-a_n) \pmod{p}$

- (Fermat) $x^{p-1} - 1 \equiv (x-1)\cdots(x-p+1) \pmod{p}$

  Wilson

- $d \mid p-1$ then $x^d - 1 \equiv 0$ has precisely d zeros

  $\because$ $\underbrace{(x^{p-1}-1)}_{p-1 \text{ zeros}} \equiv \underbrace{(x^d-1)}_{d \text{ zeros}} g(x)$

False for Composite moduli

① $m_1 \ldots m_k$ coprime in pairs

② polynomial f with integer coefficients

③ $f(x) \equiv 0 \pmod{m_j}$ has s_j solutions

$\Rightarrow$ $f(x) \equiv 0 \pmod{m_1 \cdots m_k}$ has $s_1 \cdots s_k$ solutions (CRT)

False for prime power moduli

: $x^2 \equiv 1 \pmod{8}$ has four solutions

if prime p does not divide discriminant of f

then theorem holds for all power p^j

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e.g $f(x) = x^2 - a$ with $p \nmid a$ (p is odd prime)

① $f(y) = y^2 - a \equiv 0 \pmod{p^j}$

② $f(y + p^j z) \equiv (y + p^j z)^2 - a$

$$\equiv y^2 + 2y p^j z + \cancel{p^{2j} z^2} - a$$

$$\equiv 0 \pmod{p^{j+1}}$$

③ $f(y) + 2y p^j z \equiv 0 \pmod{p^{j+1}}$

$f(y)/p^j + 2y z \equiv 0 \pmod{p}$

$(2y, p) = 1$

reduced

set of residue

6. Primitive Roots

$(a, n) = 1$ then

① $a^{\phi(n)} \equiv 1 \pmod{n}$

② order d : least $d \in \mathbb{N}$

s.t $a^d \equiv 1 \pmod{n}$

"a belongs to $d \pmod{n}$"

③ $a^k \equiv 1 \pmod{n}$ $\quad \because k = dq + r$

then $d \mid k$ i.e $d \mid \phi(n)$

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primitive root $\pmod{n}$: a that belongs to $\phi(n)$

- odd prime p
- $\psi(d)$ be # of a that belongs to $\phi(n) \pmod{p}$

$\Rightarrow$ # of primitive roots $\pmod{p}$: $\phi(p-1)$

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claim if $\psi(d) \neq 0$ then $\psi(d) = \phi(d)$

- Suppose a belongs to $d \pmod{p}$
- $x^d \equiv 1 \pmod{p}$ has precisely d solutions : $a^1, \ldots, a^d$
- does a^m belong to $d \pmod{p}$?

if $(m, d) = 1$: $d \mid d'$

if $(m, d) \neq 1$: $d \nmid d'$

$\rightsquigarrow (a^m)^{d'} \equiv 1 \pmod{p}$

when $d' = d/(m, d)$

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we know that $\sum_{d \mid p-1} \psi(d) = \sum_{d \mid p-1} \phi(d) = p-1$

Claim modulo p^j has primitive root

primitive root g $\pmod{p}$

primitive root $g' = g + px$ $\pmod{p^j}$

$g^{p-1} = 1 + pz$ with $z \equiv y + \underbrace{(p-1)}_{\text{not divisible by } p} \underbrace{g^{p-2}}\, x \pmod{p}$

$g'^{p-1} = 1 + pz$

$\hookrightarrow$ select x s.t. $(z, p) = 1$

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① g' belongs to d $\pmod{p^j}$ $\Rightarrow$ $d \mid p^{j-1}(p-1)$ by Euler

② primitive root g' $\pmod{p}$ $\Rightarrow$ $p-1 \mid d$

therefore $d = p^k (p-1)$ with $k < j$

Note. $\begin{cases} p \text{ is odd} \\ (z, p) = 1 \end{cases}$ then $(1 + pz)^{p^k} = 1 + p^{k+1} z_k$

with $(z_k, p) = 1$ $(k \geq 0)$

$\therefore$ $(1 + p^{k+1} z_k)^p = 1 + p^{k+2} z_k + \frac{p-1}{2} \cdot p^{2k+3} z_k^2 + \cdots$

$g'^d \equiv (g'^{p-1})^{p^k}$ then $k+1 = j$

$\equiv (1 + pz)^{p^k}$ $d = \phi(p^j)$

$\equiv 1 + p^{k+1} z_k$

$\equiv 1 \pmod{p^j}$

primitive roots (mod n) iff $n = 2, 4, p^j, 2p^j$

- $1 \pmod 2$, $3 \pmod 4$
- $g \pmod{p^j}$ $\Rightarrow$ odd element of g, $g+p^j \pmod{2p^j}$

$\because$ $\phi(p^j) = \phi(2p^j)$

$= \text{order of } \underbrace{G}_{\text{홀수}}$

$(n_1, n_2) = 1$ with $n_1 > 2$, $n_2 > 2$ then no primitive root (mod $\underbrace{n_1 n_2}_{= n}$)

because $a^{\frac{1}{2}\phi(n)} \equiv 1 \pmod n$

- $\phi(n_1)$ and $\phi(n_2)$ are even
- $a^{\frac{1}{2}\phi(n)} \equiv (a^{\phi(n_1)})^{\frac{1}{2}\phi(n_2)} \equiv 1 \pmod{n_1}$

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no primitive root (mod 2^j) $j > 2$

because $a^{2^{j-2}} \equiv 1 \pmod{2^j}$

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7. Indices

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primitive root $g \pmod{n}$

$\Rightarrow$ $g^1, g^2, \ldots, g^{\phi(n)}$ becomes reduced set of residues $\pmod{n}$

$\Rightarrow$ $(a, n) = 1$ iff $a \equiv g^l$

l is called "index of a with respect to g"

- $\text{ind } 1 = 0$
- $\text{ind } g = 1$
- $\text{ind } a + \text{ind } b \equiv \text{ind } (ab) \pmod{\phi(n)}$
- $\text{ind } (a^m) = m \text{ ind } a \pmod{\phi(n)}$

$\text{ind } (-1) = \frac{1}{2}\phi(n)$ $\because$

- $g^{2\,\text{ind}(-1)} \equiv 1 \pmod{n}$
- $0 < \underbrace{2\,\text{ind}(-1)}_{= \phi(n)} < 2\phi(n)$

$x^n \equiv a \pmod{p}$ if $(n, p-1) = 1$

p: prime

there exists only one solution

$n \text{ ind}(x) \equiv \text{ind}(a) \pmod{p-1}$

$x^5 \equiv 2 \pmod{7}$.

① primitive root 3

② $5 \text{ ind } x \equiv \text{ind } 2 \pmod{6}$

$\equiv 2$

③ $\text{ind } x \equiv 4 \pmod{6}$

④ $x \equiv 3^4 \equiv 4 \pmod{7}$

- no primitive root $(\text{mod } 2^j)$ $j > 0$

- however, ① 5 belongs to 2^{j-2} $(\text{mod } 2^j)$

  ② odd integer $a \equiv (-1)^{\ell} 5^{m}$ $(\text{mod } 2^j)$

$\Rightarrow$ pair (ℓ, m) has similar properties

to the index defined above.